

RIDHIMA SRIVASTAVA — 365-DAY MASTER STUDY TRACKER

2nd Year B.Tech Computer Science (ABES) | Target: October 2027 Cybersecurity Internship

Daily Schedule: 2 Hours DSA (C++) + 1 Hour Cybersecurity (3 Hours Total Daily)

MONTH 1: C++ BASICS & NETWORK FUNDAMENTALS (DAYS 1 - 30)

Day 1	Video 003: C++ Intro & Input Operators	OSI Model Overview: All 7 Layers
Day 2	Video 004: Conditionals Part 1 (if/else)	OSI Model: Layers 1 to 4 (Physical to Transport)
Day 3	Video 005: Conditionals Part 2 (Nested/Switch)	TCP vs UDP Protocols & Header structures
Day 4	Video 006: Loops Part 1 (for/while)	IP Addressing: IPv4/IPv6 & Subnetting
Day 5	Video 007: Loops Part 2 (Factorial & Prime)	MAC Addresses & Address Resolution Protocol (ARP)
Day 6	Video 008: Pattern Printing Part 1	Port Numbers & Common Services (21, 22, 80, 443)
Day 7	Video 009: Pattern Printing Part 2	Domain Name System (DNS) Resolution Steps
Day 8	Video 010: Fundamentals Live Session	Wireshark Overview & Interface Setup
Day 9	Video 011: Functions & Pass by Value/Ref	Wireshark Packet Capture: Inspect HTTP traffic
Day 10	Video 012: Pointers & Memory Addresses	Wireshark Display Filters (http, ip.src)
Day 11	Video 013: 1D Arrays Part 1 (Traversal)	Information Security CIA Triad Overview
Day 12	Video 014: 1D Arrays Part 2 (Vectors)	Confidentiality & Symmetric Encryption
Day 13	Video 015: 1D Arrays Part 3 (Two Pointers)	Integrity & Cryptographic Hashing Concepts
Day 14	LeetCode #1: Two Sum & #26: Duplicates	Availability & DDoS Attack Vectors
Day 15	LeetCode #121: Best Time to Buy/Sell Stock	TryHackMe: Pre-Security Room 1 (Networks)
Day 16	Video 016: 2D Arrays Part 1 (Matrices)	TryHackMe: Pre-Security Room 2 (Web Works)
Day 17	Video 017: 2D Arrays Part 2 (Transpose)	HTTP Request Methods (GET, POST, PUT, DELETE)
Day 18	Video 018: 2D Arrays Part 3 (Spiral Order)	HTTP Response Status Codes (200, 301, 404, 500)
Day 19	Video 019: Strings Part 1 (std::string)	Cookies, Sessions & LocalStorage Security
Day 20	Video 020: Strings Part 2 (Valid Anagrams)	TryHackMe: Pre-Security Room 3 (Web Apps)
Day 21	Video 021: 2D Arrays Live Class	OWASP Top 10 Security Overview
Day 22	Video 022: Time Complexity (Big-O)	OWASP #1: Broken Access Control Overview
Day 23	Video 023: Space Complexity Analysis	OWASP #2: Cryptographic Failures Overview
Day 24	Video 024: Bubble Sorting $O(N^2)$	OWASP #3: Injection Attacks Overview
Day 25	Video 025: Selection & Insertion Sort	SQL Injection (SQLi) Theory & Syntax
Day 26	Video 026: Complexity Live Session	PortSwigger Academy: SQL Injection Lab 1
Day 27	Video 027: Problems on Sorting	PortSwigger Academy: SQL Injection Lab 2
Day 28	Video 028: Searching Part 1 (Binary Search)	Cross-Site Scripting (XSS) Overview
Day 29	Video 029: Searching Part 2 (Lower/Upper)	PortSwigger Academy: XSS Lab 1
Day 30	Video 030: Searching Part 3 (Answer Space)	Month 1 Review & TryHackMe Room Completion

MONTH 2: RECURSION, ADVANCED SORTING & OWASP SECURITY (DAYS 31 - 60)

Day 31	Video 031: Recursion Part 1 (Base Cases)	PortSwigger Academy: XSS Lab 2
Day 32	Video 032: Recursion Part 2 (Fibonacci)	PortSwigger Academy: XSS Lab 3
Day 33	Video 033: Recursion Part 3 (Print Subsets)	PortSwigger Academy: XSS Lab 4
Day 34	Video 034: Sorting Live Session	Cross-Site Request Forgery (CSRF)
Day 35	Video 035: Recursion Part 4 (Backtracking)	PortSwigger Academy: CSRF Lab 1
Day 36	Video 036: Advance Sorting 1 (Merge Sort)	TryHackMe: Web Fundamentals Room 1
Day 37	Video 037: Advance Sorting 2 (Quick Sort)	TryHackMe: Web Fundamentals Room 2
Day 38	Video 038: Recursion Live Session	TryHackMe: Web Fundamentals Room 3
Day 39	Video 039: Advance Sorting 3 (Cyclic Sort)	TryHackMe: Web Fundamentals Room 4
Day 40	Video 040: Special Algos 1 (Prefix Sum)	TryHackMe: Web Fundamentals Room 5
Day 41	Video 041: Special Algos 2 (Sliding Window)	TryHackMe: Web Fundamentals Room 6
Day 42	Video 042: Special Algos 3 (Sieve)	Deloitte Cyber Job Simulation Task 1
Day 43	Video 072: Greedy Algorithms Part 2	Deloitte Cyber Job Simulation Task 2
Day 44	Video 073: Greedy Algorithms Part 3	Deloitte Cyber Job Simulation Task 3
Day 45	Video 074: DP 1 Part 1 (Memoization)	OverTheWire: Bandit Level 0 to Level 2
Day 46	Video 075: DP 1 Part 2 (Tabulation)	OverTheWire: Bandit Level 3 to Level 5
Day 47	Video 076: DP 1 Part 3 (1D DP)	OverTheWire: Bandit Level 6 to Level 8
Day 48	LeetCode #53: Maximum Subarray	OverTheWire: Bandit Level 9 to Level 10
Day 49	LeetCode #15: 3Sum (Two Pointers)	Linux User Management (useradd, usermod)
Day 50	LeetCode #11: Container With Most Water	Linux Group Permissions (chmod, chown)
Day 51	LeetCode #42: Trapping Rain Water	Linux File System Hierarchy (/etc, /var)
Day 52	LeetCode #3: Longest Substring	Shell Scripting Basics: Bash Variables
Day 53	LeetCode #76: Minimum Window Substring	Shell Scripting: Automated Log Search
Day 54	LeetCode #56: Merge Intervals	Command Line Recon (whois, dig, nslookup)
Day 55	LeetCode #49: Group Anagrams	Passive Information Gathering Concepts
Day 56	Striver's Sheet: Arrays Medium Part 1	Active Information Gathering Concepts
Day 57	Striver's Sheet: Arrays Medium Part 2	PortSwigger Auth Vulnerabilities 1
Day 58	Striver's Sheet: Two Pointers Problems	PortSwigger Auth Vulnerabilities 2
Day 59	Striver's Sheet: Sliding Window Problems	Month 2 Security Review & Lab Docs
Day 60	Month 2 DSA Revision & Code Clean Up	Month 2 Cybersecurity Assessment

MONTH 3: LINKED LISTS, STACKS, QUEUES & LINUX SECURITY (DAYS 61 - 90)

Day 61	Singly Linked List: Struct Node & Head	OverTheWire: Bandit Level 11 (Base64)
Day 62	Singly Linked List: Insertion Methods	OverTheWire: Bandit Level 12 (HexDump)

Day 63	Singly Linked List: Deletion Methods
Day 64	LeetCode #206: Reverse a Linked List
Day 65	LeetCode #876: Middle of Linked List
Day 66	LeetCode #141: Cycle Detection (Floyd's)
Day 67	LeetCode #142: Linked List Cycle II
Day 68	LeetCode #21: Merge Two Sorted Lists
Day 69	Doubly Linked List Implementation
Day 70	Circular Linked List Implementation
Day 71	Stacks: Array & Linked List Build
Day 72	Stacks: C++ STL (std::stack)
Day 73	LeetCode #20: Valid Parentheses
Day 74	LeetCode #155: Min Stack Design O(1)
Day 75	Infix, Prefix & Postfix Expressions
Day 76	Postfix Expression Evaluation Stack
Day 77	Monotonic Stack: Next Greater Element I
Day 78	Monotonic Stack: Next Greater Element II
Day 79	LeetCode #739: Daily Temperatures
Day 80	LeetCode #84: Largest Histogram Area
Day 81	Queues: Array & Linked List Build
Day 82	C++ Queue STL (std::queue)
Day 83	Double Ended Queue (std::deque)
Day 84	LeetCode #225: Implement Stack via Queue
Day 85	LeetCode #232: Implement Queue via Stack
Day 86	LeetCode #239: Sliding Window Max
Day 87	Linked List & Stack Revision
Day 88	Striver's Sheet: Linked List Mediums
Day 89	Striver's Sheet: Stack & Queue Mediums
Day 90	Month 3 Revision & Code Submission

MONTH 4 TO MONTH 12 COMPLETE DAILY MILESTONES ROADMAP

Days 91-120	Month 4: Binary Search 1D/2D & Answer Space (LeetCode #33, #875, #1011)
Days 121-150	Month 5: Hash Maps, Binary Trees, BST & Trie (LeetCode #1, #98, #208)
Days 151-180	Month 6: Graph BFS/DFS, Dijkstra, Topological Sort (LeetCode #200, #207)
Days 181-210	Month 7: 1D & 2D Dynamic Programming (LeetCode #70, #198, #322, #72)
Days 211-240	Month 8: Advanced LeetCode Sprint (Arrays, Trees, Graphs, DP)
Days 241-270	Month 9: Portfolio Project 1 — C++ Encrypted File Vault CLI Tool
Days 271-300	Month 10: Portfolio Project 2 — C++ / Python Network Packet Sniffer
Days 301-330	Month 11: LeetCode Top 150 Interview Questions Sprint
Days 331-365	Month 12: Resume PDF Launch & October 2027 Internship Hiring Sprint

OverTheWire: Bandit Level 13 (SSH Keys)
OverTheWire: Bandit Level 14 (Netcat)
OverTheWire: Bandit Level 15 (SSL Netcat)
OverTheWire: Bandit Level 16 (Port Scan)
OverTheWire: Bandit Level 17 (Diff Keys)
OverTheWire: Bandit Level 18 (Shell Bypass)
OverTheWire: Bandit Level 19 (SetUID)
OverTheWire: Bandit Level 20 (Sockets)
Linux Process Management (ps, htop, kill)
Linux Signals (SIGINT, SIGTERM, SIGKILL)
Linux Service Management (systemctl)
Linux System Logs (journalctl -u)
Linux Auth Logs (/var/log/auth.log)
TryHackMe: Linux Fundamentals Part 1
TryHackMe: Linux Fundamentals Part 1
TryHackMe: Linux Fundamentals Part 2
TryHackMe: Linux Fundamentals Part 2
TryHackMe: Linux Fundamentals Part 3
TryHackMe: Linux Fundamentals Part 3
Cron Security: Insecure Cron Permissions
SUID / SGID Insecure Executables
SUID Privilege Escalation Fundamentals
Linux Firewall Basics (ufw, iptables)
Port Forwarding & SSH Tunnelling
Network Access Control Lists (NACLs)
TryHackMe: Passive Reconnaissance Room
TryHackMe: Active Reconnaissance Room
Month 3 Security Assessment

Nmap Network Reconnaissance (-sS, -sV, NSE Scripts)
Cryptography: AES-256, RSA, SHA-256 (CryptoHack)
SIEM Log Monitoring, Splunk SPL & SOC Level 1
Vulnerability Scanners: Nessus & OpenVAS Auditing
Web Pentesting with Burp Suite (PortSwigger Labs)
OpenSSL C++ Integration & AES-256 Security Audit
Packet Parsing (IP/TCP) & SYN Flood Anomaly Detection
CompTIA Security+ SY0-701 & Google Cyber Certificate
Daily Applications to 5 Cyber Analyst & SOC Intern Roles